SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0706 **COURSE NAME:** COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I **M Nishanth (192511154)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M Nishanth

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X3+1.

   i.) Use polynomial long division to determine the message that should be transmitted.

   ii.) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

   b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

   b)A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps highspeed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – CSA0713 COMPUTER NETWORKS (CO2)

Annexure A – Analytical Problem Solving: Worked Solutions

Question 1: CRC Error Detection

Given message: 11100011, Generator polynomial: $X^3 + 1$, which in binary form (degree 3, 4 bits) is 1001.

(i) Transmitted Message using Polynomial Long Division

Since the generator polynomial has degree 3, three zero bits are appended to the message before division:

Augmented message = 11100011 + 000 = 11100011000

Performing modulo-2 (XOR) polynomial long division of 11100011000 by 1001:

11100011000 ÷ 1001 → Remainder (CRC) = 100

The 3-bit CRC remainder obtained is 100. This remainder is appended to the original message in place of the appended zeros to form the transmitted codeword:

Transmitted message = 11100011 (data) + 100 (CRC) = 11100011100

Verification: Dividing 11100011100 by 1001 gives a remainder of 000, confirming the codeword is valid and correctly generated.

(ii) Effect of Inverting the Leftmost Bit

If the leftmost bit of the transmitted codeword 11100011100 is inverted (1 → 0), the received bit stream becomes:

Received message = 01100011100

The receiver divides the received bit stream by the same generator polynomial 1001. Carrying out the modulo-2 division:

01100011100 ÷ 1001 → Remainder = 010

Since the remainder is 010 (non-zero), the receiver concludes that the received data does not form a valid codeword. A valid, error-free codeword must always produce a remainder of 000 when divided by the generator polynomial. Because the computed remainder is non-zero, the receiver detects that a transmission error has occurred and can request retransmission of the frame.

Question 2

(a) Throughput Calculation

Given: Bandwidth = 10 Mbps, Average frames = 12,000 frames/minute, Frame size = 10,000 bits/frame.

Total bits transmitted per minute = 12,000 × 10,000 = 1,20,000,000 bits/minute

Throughput = 1,20,000,000 / 60 seconds = 20,00,000 bps = 2 Mbps

Interpretation: Although the network bandwidth (capacity) is 10 Mbps, the actual achieved throughput is only 2 Mbps. This shows that throughput (the actual data delivery rate) is often much lower than the theoretical bandwidth due to overheads, frame processing, and network conditions.

(b) Channel Capacity (Shannon's Theorem)

Given: Bandwidth B = 3000 Hz (300 Hz to 3300 Hz), Signal-to-Noise Ratio (SNR) = 3162.

Shannon Capacity: $C = B \times \log_2(1 + SNR)$

$C = 3000 \times \log_2(1 + 3162) = 3000 \times \log_2(3163)$

$\log_2(3163) \approx 11.63$

$C \approx 3000 \times 11.63 \approx 34{,}890$ bps ≈ 34.89 kbps

Interpretation: The theoretical maximum channel capacity of this noisy channel, per Shannon's theorem, is approximately 34.89 kbps. This is the upper limit on error-free data rate achievable over this channel given its bandwidth and noise level.

Question 3: Hamming (12, 8) Error-Correcting Code

In the Hamming(12,8) code, 8 data bits are protected by 4 redundant (parity) bits, giving a total codeword length of 12 bits. The number of parity bits r must satisfy $2^r \geq m + r + 1$, where m = 8 data bits. For r = 4: $2^4 = 16 \geq 8 + 4 + 1 = 13$, which holds true, confirming 4 parity bits are sufficient.

Parity bits are placed at positions that are powers of 2 (positions 1, 2, 4, 8), and data bits occupy the remaining positions (3, 5, 6, 7, 9, 10, 11, 12). Each parity bit checks a specific set of positions determined by the binary representation of the position numbers:

• P1 (position 1) checks positions: 1, 3, 5, 7, 9, 11

• P2 (position 2) checks positions: 2, 3, 6, 7, 10, 11

• P4 (position 4) checks positions: 4, 5, 6, 7, 12

• P8 (position 8) checks positions: 8, 9, 10, 11, 12

<u>Illustrative Example</u>

Consider an 8-bit patient-data sample: d1 d2 d3 d4 d5 d6 d7 d8 = 1 0 1 1 0 0 1 0.

Placing data bits into positions 3, 5, 6, 7, 9, 10, 11, 12 respectively and computing even parity for each parity bit (XOR of the covered positions) gives the following 12-bit transmitted codeword:

Position : 1 2 3 4 5 6 7 8 9 10 11 12

Bit : 1 0 1 0 0 1 1 1 0 0 1 0

(P1=1, P2=0, d1=1, P4=0, d2=0, d3=1, d4=1, P8=1, d5=0, d6=0, d7=1, d8=0)

Suppose electromagnetic interference flips the bit at position 7 during wireless transmission (1 → 0). The received codeword is: 1 0 1 0 0 1 0 1 0 0 1 0.

At the receiver, the four parity checks are recomputed by XOR-ing each parity bit with its covered data positions. The results are combined into a syndrome word (c8 c4 c2 c1):

$c1 = 1$, $c2 = 1$, $c4 = 1$, $c8 = 0$ → Syndrome = 0111 (binary) = 7 (decimal)

The non-zero syndrome value 7 directly indicates that the bit at position 7 is in error. The receiver simply inverts the bit at position 7 to recover the original, correct codeword – without needing any retransmission.

Relevance to the scenario: This is exactly why the hospital chose Hamming(12,8): it allows the monitoring station to automatically detect and correct single-bit errors caused by wireless interference in real time, ensuring doctors receive accurate heart rate and blood pressure readings instantly, without the delay of requesting a retransmission.

Question 4

<u>(a) IPv4 Subnetting</u>

Given network: 172.16.10.0/24, subnetted using a /27 mask.

Bits borrowed for subnetting = 27 − 24 = 3 bits

Number of subnets = $2^3 = 8$ subnets

Host bits remaining = $32 - 27 = 5$ bits

Usable hosts per subnet = $2^5 - 2 = 30$ usable hosts

Each subnet block size = $2^5 = 32$ addresses. The 8 subnets are:

Subnet	Network Address	Usable Host Range	Broadcast Address
1	172.16.10.0/27	.1 – .30	172.16.10.31
2	172.16.10.32/27	.33 – .62	172.16.10.63
3	172.16.10.64/27	.65 – .94	172.16.10.95
4	172.16.10.96/27	.97 – .126	172.16.10.127
5	172.16.10.128/27	.129 – .158	172.16.10.159
6	172.16.10.160/27	.161 – .190	172.16.10.191
7	172.16.10.192/27	.193 – .222	172.16.10.223
8	172.16.10.224/27	.225 – .254	172.16.10.255

The address 172.16.10.78 falls between 64 and 95, so it belongs to Subnet 3: 172.16.10.64/27.

Network address of this subnet = 172.16.10.64

Broadcast address of this subnet = 172.16.10.95

The address 172.16.10.95 is the broadcast address of the same subnet (172.16.10.64/27), so it does fall within the same subnet's address range – though as the broadcast address it cannot be assigned to a host.

(b) IPv6 Addressing

Given address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed (shortest) form = 2001:db8::ff00:42:8329

Rule applied: leading zeros within each 16-bit group are removed, and the single longest run of consecutive all-zero groups is replaced once with '::' .

Expanded (full) form = 2001:0db8:0000:0000:0000:ff00:0042:8329

Using a /64 prefix length, the network prefix is the first 64 bits (first 4 groups):

Network prefix = 2001:0db8:0000:0000 /64 (i.e., 2001:db8::/64)

Number of possible host addresses with a /64 prefix:

Host bits = 128 − 64 = 64 bits

Total possible host addresses = $2^{64} \approx 1.844 \times 10^{19}$ addresses

Question 5: Flow Control Problem During Online Examination

Primary Networking Problem Identified

The scenario describes a classic flow-control problem. The examination server (connected via a 10 Gbps link) transmits data far faster than many student devices, especially those on slow Wi-Fi or mobile networks (5–20 Mbps), can receive and process it. This is not a bandwidth or congestion problem alone – it is fundamentally a mismatch between sender speed and receiver processing capacity, which is precisely the problem that flow control mechanisms are designed to solve.

How the Speed Mismatch Affects Communication

1. Receiver buffer overflow: Slower student devices have limited buffer space. When the server keeps sending data at high speed without regard to the receiver's consumption rate, incoming packets arrive faster than the device can process and clear its buffer, causing the buffer to overflow and packets to be dropped.

2. Packet loss and retransmissions: Once packets are dropped due to buffer overflow, the receiver cannot acknowledge them. This triggers timeout-based or duplicate-acknowledgment-based retransmissions from the server, which adds further load onto already congested slow links.

3. Delayed acknowledgments and freezing: As the receiver struggle to keep up, acknowledgments are delayed, causing the sender's congestion/flow window to behave inefficiently, resulting in stalled or 'frozen' data transfer from the student's perspective.

4. Increased network traffic: Repeated retransmissions of already-sent data multiply the effective traffic on the network, worsening delays for all students, not just the ones with slow connections.